

SAFETY DATA SHEET

ECOLAB®

ULTRASIL 10

1. Chemical product and company identification

Chinese name : 奇疏10
Product name : ULTRASIL 10
Other names : None.
Product code : 987388
Recommended use and restrictions : Cleaning product
Use only for the purpose on the product label.
Manufacturer/Supplier : Ecolab Limited
Address/Telephone : 6F, No. 13, Sec. 2, Peitou Rd., Peitou Dist.,
Taipei, 112, Taiwan, R.O.C.
(02) 2898-4800
Emergency telephone number : 1-651-222-5352 (in USA)

2. Hazards identification

GHS Classification : CORROSIVE TO METALS - Category 1
ACUTE TOXICITY: ORAL - Category 5
SKIN CORROSION/IRRITATION - Category 1A
SERIOUS EYE DAMAGE/ EYE IRRITATION - Category 1
AQUATIC TOXICITY (ACUTE) - Category 3

GHS label elements

Symbol

:

**Signal word**

: Danger

Hazard statements

: May be corrosive to metals.
May be harmful if swallowed.
Causes severe skin burns and eye damage.
Harmful to aquatic life.

Precautionary statements

Prevention

: Wear protective gloves. Wear eye or face protection. Wear protective clothing. Keep only in original container. Avoid release to the environment. Do not breathe dust or mist. Wash hands thoroughly after handling.

Response

: Absorb spillage to prevent material damage. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER or physician. IF SWALLOWED: Immediately call a POISON CENTER or physician. Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. Wash contaminated clothing before reuse. Immediately call a POISON CENTER or physician. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or physician.

Other hazards

: Handling and/or processing of this material may generate a dust which can cause mechanical irritation of the eyes, skin, nose and throat.

3. Composition/information on ingredients

Nature of material : Mixture
Chemical properties : Cleaning product

Hazardous ingredients	Concentration Range (%)	CAS number
acetic acid, (ethylenedinitrilo)tetra-, tetrasodium salt	20 - 50	64-02-8
sodium hydroxide	5 - 20	1310-73-2
triposphoric acid, pentasodium salt	5 - 20	7758-29-4
sodium dodecylbenzene sulfonate	1 - 5	25155-30-0

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

4. First aid measures

First aid measures

- Eye contact** : Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
- Skin contact** : Take off immediately all contaminated clothing. Rinse skin with water or shower. Get medical attention immediately. Wash clothing before reuse. Clean shoes thoroughly before reuse.
- Inhalation** : Remove victim to fresh air and keep at rest in a position comfortable for breathing. Get medical attention immediately.
- Ingestion** : Get medical attention immediately. Rinse mouth. Do not induce vomiting.

Most important symptoms/effects, acute and delayed

- Eye contact** : Causes serious eye damage.
- Skin contact** : Causes severe burns.
- Inhalation** : May cause respiratory irritation.
- Ingestion** : May be harmful if swallowed. May cause burns to mouth, throat and stomach.

- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. If ventilation is inadequate, use respirator that will protect against dust/mist. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.
- Notes to physician** : In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.

See toxicological information (Section 11)

5. Fire-fighting measures

- Suitable fire extinguishing media** : Use dry chemical powder.
- Specific hazards arising from the chemical** : Fine dust clouds may form explosive mixtures with air. This material is harmful to aquatic life. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.
- Hazardous thermal decomposition products** : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
nitrogen oxides
sulfur oxides
phosphorus oxides
metal oxide/oxides
- Specific fire-fighting methods** : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Move containers from fire area if this can be done without risk. Use water spray to keep fire-exposed containers cool.

5. Fire-fighting measures

Special protective equipment for fire-fighters : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

6. Accidental release measures

Personal precautions : Initiate company's spill response procedures immediately. Keep people out of area. Put on appropriate personal protective equipment (see section 8). Do not touch or walk through spilled material.

Environmental precautions : Avoid contact of spilled material and runoff with soil and surface waterways.

Methods for cleaning up : If emergency personnel are not present, contain and collect small spillages with non-combustible absorbent material, e.g. sand, earth or other suitable material and place in a suitable container for disposal. Use a water rinse for final clean-up.

7. Handling and storage

Handling : Do not ingest. Do not get in eyes or on skin or clothing. Do not breathe dust. Use only with adequate ventilation. Wash thoroughly after handling.

Storage : Keep out of reach of children. Keep container tightly closed.

Store between the following temperatures: -30 and 40°C

8. Exposure controls/personal protection

Control parameters

Ingredient name	Exposure limits
sodium hydroxide	Taiwan Council of Labor Affairs (Taiwan, 1/2010). STEL: 4 mg/m ³ 15 minutes. TWA: 2 mg/m ³ 8 hours.

Biological indicator : Not available.

Appropriate engineering controls : Use only with adequate ventilation. If user operations generate dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Personal protection

Hygiene measures : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing.

Eye protection : Use chemical splash goggles. For continued or severe exposure wear a face shield over the goggles.

Hand protection : Use chemical-resistant, impervious gloves.

Skin protection : Use synthetic apron, other protective equipment as necessary to prevent skin contact.

Respiratory protection : A respirator is not needed under normal and intended conditions of product use.

9. Physical and chemical properties

Appearance

Physical state : Powder

Color : White with colored particles

Odor : Faint odor

pH : 12.4 to 12.8 (1%)

Melting point : Not available.

9. Physical and chemical properties

Flash point	: > 100°C Product does not support combustion.
Evaporation rate (butyl acetate = 1)	: Not available.
Flammability (solid, gas)	: Not available.
Explosion limits	: Not available.
Vapor pressure	: Not available.
Vapor density	: Not available.
Relative density	: Weighted average: 1.35 (Water = 1)
Solubility	: Easily soluble in the following materials: cold water and hot water.
Partition coefficient: n-octanol/water	: Not available.
Auto-ignition temperature	: Not available.

10. Stability and reactivity

Stability	: The product is stable.
Possibility of hazardous reactions	: Under normal conditions of storage and use, hazardous reactions will not occur.
Conditions to avoid	: Avoid the creation of dust when handling and avoid all possible sources of ignition (spark or flame). Take precautionary measures against electrostatic discharges. To avoid fire or explosion, dissipate static electricity during transfer by grounding and bonding containers and equipment before transferring material. Prevent dust accumulation.
Materials to avoid	: Extremely reactive or incompatible with the following materials: acids. Reactive or incompatible with the following materials: metals and moisture.
Hazardous decomposition products	: Under normal conditions of storage and use, hazardous decomposition products should not be produced.

11. Toxicological information

Route of exposure	: Skin contact, Eye contact, Ingestion
Symptoms	
Eye contact	: Adverse symptoms may include the following: pain watering redness
Skin contact	: Adverse symptoms may include the following: pain or irritation redness blistering may occur
Inhalation	: Adverse symptoms may include the following: respiratory tract irritation coughing Respiratory tract irritation
Ingestion	: Adverse symptoms may include the following: stomach pains

Acute toxicity

Eye contact	: Causes serious eye damage.
Skin contact	: Causes severe burns.
Inhalation	: May cause respiratory irritation.
Ingestion	: May be harmful if swallowed. May cause burns to mouth, throat and stomach.

Toxicity data

11. Toxicological information

Product/ingredient name	Result	Species	Dose	Exposure
acetic acid, (ethylenedinitrilo)tetra-, tetrasodium salt	LD50 Oral	Rat	1700 mg/kg	-
triposphoric acid, pentasodium salt	LD50 Dermal	Rabbit	>4650 mg/kg	-
	LD50 Oral	Rat	3050 mg/kg	-
sodium dodecylbenzene sulfonate	LD50 Dermal	Rat	>2000 mg/kg	-
	LD50 Oral	Rat	1086 mg/kg	-

Chronic toxicity

Carcinogenicity	: No known significant effects or critical hazards.
Mutagenicity	: No known significant effects or critical hazards.
Teratogenicity	: No known significant effects or critical hazards.
Developmental effects	: No known significant effects or critical hazards.
Fertility effects	: No known significant effects or critical hazards.

12. Ecological information

Ecotoxicity : This material is harmful to aquatic life.

Aquatic and terrestrial toxicity

Product/ingredient name	Test	Result	Species	Exposure
acetic acid, (ethylenedinitrilo)tetra-, tetrasodium salt	-	Acute LC50 121 mg/l	Fish	96 hours
sodium hydroxide	-	Acute EC50 40 mg/l	Daphnia	48 hours
triposphoric acid, pentasodium salt	-	Acute EC50 >100 mg/l	Daphnia	48 hours
sodium dodecylbenzene sulfonate	-	Acute LC50 3.2 mg/l	Fish	96 hours

Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
Not available.				

<u>Product/ingredient name</u>	<u>Aquatic half-life</u>	<u>Photolysis</u>	<u>Biodegradability</u>
Not available.			

Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
acetic acid, (ethylenedinitrilo)tetra-, tetrasodium salt	5.01	-	high
sodium dodecylbenzene sulfonate	1.96	-	low

Mobility in soil : Not available.

Other adverse effects : No known significant effects or critical hazards.

13. Disposal considerations

Disposal methods : The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

14. Transport information

Certain shipping modes or package sizes may have exceptions from the transport regulations. The classification provided may not reflect those exceptions and may not apply to all shipping modes or package sizes.

UN number : UN1823

Proper shipping name : SODIUM HYDROXIDE, SOLID

Class : 8

Packing group : II

Marine pollutant (Yes / No) : Not a pollutant.

Specific precautionary transport measures and conditions : -

See shipping documents for specific transportation information.

15. Regulatory information

TCSCA List of toxic chemicals

<u>Listed no.</u>	<u>Series no.</u>	<u>Product name</u>	<u>RQ</u>	<u>Class</u>
Not available.				

List of chemicals for which manufacturing or handling is defined as "work specially hazardous to health" : Not available.

List of chemicals reputed to be a "threat of imminent danger" : This product contains substances considered to be a "Threat of imminent danger": sodium hydroxide.

LSHL Article 21 : Not available.

Safety, health and environmental regulations specific for the product : Not available.

References : -Regulation of Labeling and Hazard Communication of Dangerous and Harmful Materials.
-Labor Safety and Health Act.

16. Other information

History

Date of issue : 2013 . 12 . 5

Date of previous issue : 2012 . 12 . 7

Organization that prepared the SDS : ECOLAB

Person who prepared the SDS : Regulatory Affairs

Notice to reader

The above information is believed to be correct with respect to the formula used to manufacture the product in the country of origin. As data, standards, and regulations change, and conditions of use and handling are beyond our control, NO WARRANTY, EXPRESS OR IMPLIED, IS MADE AS TO THE COMPLETENESS OR CONTINUING ACCURACY OF THIS INFORMATION.